Learnable Path-Complex Features and Unified Path–Cellular Message Passing for Link Prediction

Farhan Sadeek

Dartmouth College

April 2026

Abstract

Path Complex Networks (PCN) generalize simplicial and cellular message passing by lifting a graph to its path complex and exchanging messages between k-paths through boundary, coboundary, and upper/lower-adjacency relations. In the original formulation, the feature of a k-cell $(v_0, \ldots, v_k)$ is a deterministic scatter-sum (or scatter-mean) of its constituent node features, discarding both the *order* along the path and any learnable capacity at the lifting step. Moreover, PCN commits to a single topological lift—paths, cliques, or rings—so the model cannot simultaneously exploit the complementary inductive biases offered by different cell types. We address both limitations. First, we introduce a position-aware learnable feature initialization in which each position along a k-cell receives its own linear (or MLP) map, yielding order-sensitive path features at no asymptotic cost. Second, we construct a unified path + cellular complex in which path-cells and ring-cells coexist at the same dimension and communicate through shared edge boundaries, so that a triangle ABC is simultaneously a path $A{\to}B{\to}C$ and a 3-cycle without duplicating adjacency bookkeeping. We evaluate both contributions on Cora link prediction. Our experiments ablate (i) deterministic vs. learnable initialization and (ii) path-only vs. unified complexes, and we report test AUC across 10 seeds for every configuration.

1 Introduction

Graph neural networks that rely on the 1-Weisfeiler–Lehman (1-WL) update rule are fundamentally limited in their ability to distinguish non-isomorphic graphs [Xu et al., 2019]. Higher-order topological GNNs escape this limit by lifting the graph to a richer combinatorial object—a simplicial complex [Bodnar et al., 2021a], a regular cell complex [Bodnar et al., 2021b], or a path complex [Truong and Chin, 2024]—and running message passing over its cells. Each lift carries a different inductive bias: simplicial complexes privilege cliques, cell complexes privilege induced cycles, and path complexes privilege simple paths and thus make no structural assumption at all.

PCN [Truong and Chin, 2024] is attractive precisely because paths exist in every graph, but the published implementation leaves two degrees of freedom unused. (1) Cell features at dimension $k \geq 1$ are initialized by a deterministic scatter-sum of node features. Because summation is symmetric, the feature of the path $A{\to}B{\to}C$ is identical to the feature of $C{\to}B{\to}A$, which is counterintuitive for an object whose defining property is an ordering of its vertices. (2) The lift is monolithic: one picks paths *or* rings *or* cliques. In practice, however, the same substructure—say a triangle—is a clique, a 3-cycle, and a path of length two simultaneously, and a model with access to all three views should, in principle, be strictly more expressive than any one of them.

We propose two small but targeted modifications to PCN:

1. **Learnable, position-aware cell initialization.** For a k-cell $(v_0, \dots, v_k)$ we learn $k{+}1$ linear maps $W_0, \dots, W_k$, one per position, and define $\mathbf{h}_\sigma^{(0)} = \sum_{i=0}^{k} W_i \mathbf{x}_{v_i}$. The node features now enter the path-complex features through a learned, ordered transform rather than a symmetric sum.
2. **Unified path + cellular complex.** We build a single complex in which path-cells and ring-cells coexist at dimension two, share the same edge id-map, and communicate through their shared edges during message passing. A triangle therefore sends messages *as* a ring-cell and simultaneously appears as a path-cell; the two views are fused downstream through the dim-1 edges they both bound.

Both modifications are drop-in: the message-passing layer is unchanged, and the per-cell feature dimension is unchanged. We evaluate on Cora link prediction in the transductive setting of Kipf and Welling [2016], reporting mean ± std test AUC over 10 seeds for four configurations: $(\text{path}, \text{det})$, $(\text{path}, \text{learn})$, $(\text{unified}, \text{det})$, $(\text{unified}, \text{learn})$.

2 Background

Path complexes. Given a graph $G = (V, E)$, a *simple path* of length k is an ordered tuple $(v_0, \dots, v_k)$ of distinct vertices with $\{v_i, v_{i+1}\} \in E$ for all i. The set $P_k(G)$ of all such paths, together with boundary maps ∂ defined by deletion of the leftmost or rightmost vertex, forms a *path complex* [Grigor'yan et al., 2012, Truong and Chin, 2024]. Message passing in PCN operates over $P_0(G) \cup P_1(G) \cup \cdots \cup P_K(G)$ via boundary, coboundary, and upper/lower adjacencies between paths, in direct analogy with CWN.

Deterministic feature initialization. In the public PCN implementation, the feature of a k-cell at the input layer is

$$\mathbf{h}_\sigma^{(0)} = \bigoplus_{v \in \sigma} \mathbf{x}_v, \qquad \sigma = (v_0, \dots, v_k), \tag{1}$$

where $\oplus \in \{\text{sum}, \text{mean}\}$. We refer to this as the *deterministic init*. Equation (1) is symmetric under any permutation of $(v_0, \dots, v_k)$, so path orientation is lost and cannot be recovered by subsequent layers that operate only on $\mathbf{h}_\sigma^{(0)}$.

3 Method

3.1 Learnable position-aware initialization

We replace (1) with

$$\mathbf{h}_\sigma^{(0)} = \sum_{i=0}^{k} W_i^{(k)} \, \mathbf{x}_{v_i}, \qquad \sigma = (v_0, \dots, v_k), \tag{2}$$

where $W_i^{(k)} \in \mathbb{R}^{d \times d_{\text{in}}}$ is a learnable linear map specific to the pair (k, i) of dimension and position. Asymptotically the cost is identical to (1): each node contributes exactly once to the feature of each cell it bounds, and the only change is that the contribution is multiplied by a position-specific matrix before being accumulated. When $W_i^{(k)} = W$ for all (k, i), (2) reduces to a linear projection of the scatter-sum, so the deterministic init is strictly less expressive.

In the implementation, the path-complex builder already records a `cell_node_index` tensor of shape $[2, \sum_\sigma (k_\sigma + 1)]$ and a `position_index` tensor; the learnable init iterates over positions $0, \ldots, k$ and performs a masked linear projection followed by an `index_add_` to accumulate into the cell feature buffer (`lib/layers/learnable_features.py`).

3.2 Unified path–cellular complex

Let $P_k(G)$ denote the set of simple k-paths and $R(G)$ the set of induced cycles up to length L. We build a *unified complex* $\mathcal{C}(G)$ with cells

$$\mathcal{C}_0 = V, \quad \mathcal{C}_1 = P_1(G), \quad \mathcal{C}_2 = P_2(G) \cup R(G), \quad \mathcal{C}_k = P_k(G) \text{ for } k \geq 3.$$

The dim-0 and dim-1 id-maps are shared across path-cells and ring-cells (edges are canonicalized to sorted tuples), so a ring-cell and a path-cell that share an edge point to the same edge id. Boundary extraction branches on cell type: path-cells use the left/right-deletion rule, and ring-cells use the standard CW boundary map that returns every edge of the cycle. Upper and lower adjacencies are then computed by the existing PCN routine on the combined boundary tables, so messages flow seamlessly between a ring-cell and a path-cell that happen to share an edge.

Why this helps link prediction. Link prediction rewards models whose node embeddings capture closed-neighborhood structure: a pair (u, v) is more likely to be linked when u and v participate in short cycles. Path-cells expose only open neighborhoods; ring-cells expose closed ones. Since cycles of length $\leq L$ in Cora are overwhelmingly short, adding ring-cells to a path complex injects a targeted topological signal that paths alone do not carry.

4 Experiments

4.1 Setup

We evaluate on Cora in the transductive link-prediction setting of Kipf and Welling [2016]: edges are split 85/5/10 into train/val/test with an equal number of sampled negatives per split. The path complex (or unified complex) is built from training message-passing edges only. All models use a 2-layer PCN encoder with hidden dimension 64, dropout 0.3, batch normalization, a dot-product decoder, and Adam with lr = 0.01 and `ReduceLROnPlateau` scheduling. Each run is 200 epochs with early stopping (patience 20 on val AUC). We report mean ± std over 10 seeds $(0, \ldots, 9)$.

4.2 Ablations

Table 1 reports the four configurations. The baseline (path, deterministic) achieves test AUC 0.7113 ± 0.0239, consistent with the fact that PCN was not designed for link prediction and a pure-path lift provides little closed-neighborhood signal. We compare against this baseline the three modifications introduced in Section 3.

4.3 Discussion

Neither of our modifications is expected to close the gap to specialised link-prediction architectures such as SEAL [Zhang and Chen, 2018], NBFNet, or a well-tuned VGAE—those models exploit properties of the link-prediction problem (subgraph enclosure, Bellman–Ford relational paths, bi-linear decoders on structural features) that are largely orthogonal to the topological lift studied

Complex	Init	Val AUC	Test AUC
Path	Deterministic	0.7344 ± 0.0274	0.7113 ± 0.0239
Path	Learnable	[to be filled from `pcn_cora_lp_learnable`]	[...]
Unified	Deterministic	[to be filled from `pcn_cora_lp_unified`]	[...]
Unified	Learnable	[to be filled from `pcn_cora_lp_unified_learnable`]	[...]

Table 1: Cora link-prediction AUC (mean ± std over 10 seeds). Baseline (Path, Deterministic) numbers are from the `pcn_cora_link_prediction` run at `experiments/results/Cora/runs/2026-04-05T13:11:20-...`. The remaining three rows are produced by the three scripts in `experiments/scripts/`.

here. What the ablation does isolate is the marginal effect, *within* the PCN family, of (i) learning the lift from data and (ii) admitting multiple cell types simultaneously.

A natural limitation of the unified complex is duplication: a 3-cycle ABC is now represented both as a ring-cell and as three path-cells $A{\to}B{\to}C$, $B{\to}C{\to}A$, $C{\to}A{\to}B$. We do not deduplicate, and we expect this to over-weight triangle structure; exploring learnable gating between cell types is a promising direction.

5 Conclusion

We described two small extensions to Path Complex Networks—learnable position-aware cell initialization and a unified path + cellular lift—and evaluated them on Cora link prediction. Both are implemented as drop-in modifications to the public PCN codebase, require no change to the message-passing layer, and preserve the feature budget of the original model. Future work includes extending the unified complex to admit clique-cells (simplicial) alongside paths and rings, evaluating on OGB link-prediction benchmarks, and testing whether the position-aware init offers benefits for graph-level tasks where PCN is already competitive.

References

Bodnar, C., Frasca, F., Wang, Y. G., Otter, N., Montúfar, G., Liò, P., and Bronstein, M. Weisfeiler and Lehman go topological: Message passing simplicial networks. In *ICML*, 2021.

Bodnar, C., Frasca, F., Otter, N., Wang, Y. G., Liò, P., Montúfar, G., and Bronstein, M. Weisfeiler and Lehman go cellular: CW networks. In *NeurIPS*, 2021.

Grigor'yan, A., Lin, Y., Muranov, Y., and Yau, S.-T. Homologies of path complexes and digraphs. *arXiv:1207.2834*, 2012.

Kipf, T. N. and Welling, M. Variational graph auto-encoders. In *NeurIPS Bayesian Deep Learning Workshop*, 2016.

Truong, Q. and Chin, P. Weisfeiler and Lehman go paths: Learning topological features via path complexes. In *AAAI*, 2024.

Xu, K., Hu, W., Leskovec, J., and Jegelka, S. How powerful are graph neural networks? In *ICLR*, 2019.

Zhang, M. and Chen, Y. Link prediction based on graph neural networks. In *NeurIPS*, 2018.